

In[419]:=

```
(* ::Package::*)ClearAll["Global`*"];

(*=====*)
(*BIGGER BASELINE VALIDATION-WITH APPROXIMATE LENZ TORQUE*)
(*=====*)

(*-----Physical Constants-----*)
mu0 = 4 Pi * 10^-7;
m0 = 406.0;          (*A m^2*)
freq = 50.0;
omega = 2 Pi freq;
turns = 100;
coilRadius = 0.02;
coilArea = Pi coilRadius^2;
z0 = 0.10;
Rcoil = 0.5;
Rload = 10.0;
Rtotal = Rcoil + Rload;
Lcoil = 3.0 * 10^-3;
period = 1 / freq;

(*=====*)
(*DIPOLE CONFIGURATION*)
(*=====*)
mx[t_] := m0 * Sin[omega t];
my[t_] := 0;
mz[t_] := m0 * Cos[omega t];

(*Magnetic field on z-axis from dipole*)
BzDipole[t_] := (mu0 / (4 Pi * z0^3)) * 2 * mz[t];
B0 = (mu0 / (4 Pi * z0^3)) * 2 * m0;

Print["B0 = ", B0, " T"];
Print["Peak Bz = ", B0, " T"];

(*=====*)
(*FLUX AND EMF*)
(*=====*)
flux[t_] := turns * coilArea * BzDipole[t];
emf[t_] := turns * coilArea * (mu0 / (4 Pi * z0^3)) * 2 * m0 * omega * Sin[omega t];

(*=====*)
(*RL CIRCUIT SOLUTION*)
(*=====*)
sol = NDSolve[{Lcoil current'[t] + Rtotal current[t] == emf[t], current[0] == 0},
  current, {t, 0, 0.20}, MaxStepFraction -> 1 / 1000];
```

```

Icoil = current /. sol[[1]];

(*=====*)
(*LOAD VOLTAGE AND POWER*)
(*=====*)
Vload[t_] := Rload * Icoil[t];
PowerOut[t_] := Vload[t] * Icoil[t];

(*=====*)
(*RMS VALUES*)
(*=====*)
Vrms = Sqrt[1 / 0.10 * NIntegrate[Vload[t]^2, {t, 0.10, 0.20}]];
Irms = Sqrt[1 / 0.10 * NIntegrate[Icoil[t]^2, {t, 0.10, 0.20}]];
averagePower = 1 / 0.10 * NIntegrate[PowerOut[t], {t, 0.10, 0.20}];

(*=====*)
(*METHOD 1:DIPOLE-COIL INTERACTION TORQUE (APPROXIMATE)*)
(*=====*)
(*This assumes the coil behaves as a point dipole on the z-axis*)
(*It is an APPROXIMATION,not the full electromagnetic torque*)

(*Coil magnetic moment*)
mCoil[t_] := Icoil[t] * turns * coilArea;

(*Field from coil at dipole position (on-axis approximation)*)
BcoilAtDipole[t_] := (mu0 / (4 Pi * z0^3)) * 2 * mCoil[t];

(*Torque=m_dipolexB_coil (z-component only)*)
tauDipoleCoil[t_] := -BcoilAtDipole[t] * mx[t];

Print["\n===== METHOD 1: DIPOLE-COIL INTERACTION ====="];
Print["This is an APPROXIMATE electromagnetic reaction torque"];
Print["Assumptions:"];
Print[" 1. Coil is treated as a point dipole"];
Print[" 2. Only on-axis field components are considered"];
Print[" 3. No integration over coil geometry"];

(*Average torque from Method 1*)
avgTorque1 = 1 / 0.10 * NIntegrate[tauDipoleCoil[t], {t, 0.10, 0.20}];
avgTorque1Magnitude = Abs[avgTorque1];

Print["Average Dipole-Coil Torque = ", avgTorque1Magnitude, " Nm"];

(*=====*)
(*METHOD 2:TORQUE FROM POWER (P/omega)*)
(*=====*)

```

```

torqueFromPower = averagePower / omega;

Print["\n===== METHOD 2: TORQUE FROM POWER ====="];
Print["Torque =  $P/\omega$  = ", torqueFromPower, " Nm"];
Print["This is the mechanical torque required to generate P"];

(*=====*)
(*METHOD 3:BIOT-SAVART INTEGRATION (MORE ACCURATE)*)
(*=====*)
(*Calculate the magnetic field at the dipole from the coil*)
(*Using Biot-Savart for a circular coil*)

(*Biot-Savart field on the z-axis for a circular coil*)
BcoilBiotSavart[t_] :=
  (mu0 * Icoil[t] * turns * coilRadius^2) / (2 * (coilRadius^2 + z0^2)^(3/2));

(*Torque using Biot-Savart field*)
tauBiotSavart[t_] := -BcoilBiotSavart[t] * mx[t];

Print["\n===== METHOD 3: BIOT-SAVART TORQUE ====="];
Print["This uses the actual field from a circular coil"];
Print["Formula:  $B_{coil} = (\mu_0 * I * N * R^2) / (2 * (R^2 + z_0^2)^{3/2})$ "];

(*Average torque from Method 3*)
avgTorque3 = 1 / 0.10 * NIntegrate[tauBiotSavart[t], {t, 0.10, 0.20}];
avgTorque3Magnitude = Abs[avgTorque3];

Print["Average Biot-Savart Torque = ", avgTorque3Magnitude, " Nm"];

(*=====*)
(*COMPARISON OF METHODS*)
(*=====*)
Print["\n===== COMPARISON OF TORQUE CALCULATIONS ====="];
Print["Method 1 (Dipole-Coil Approximation): ", avgTorque1Magnitude, " Nm"];
Print["Method 2 ( $P/\omega$ ): ", torqueFromPower, " Nm"];
Print["Method 3 (Biot-Savart): ", avgTorque3Magnitude, " Nm"];
Print["Ratio Method1/Method3: ", avgTorque1Magnitude / avgTorque3Magnitude];
Print["Ratio Method2/Method3: ", torqueFromPower / avgTorque3Magnitude];

(*=====*)
(*NUMERICAL PEAKS*)
(*=====*)
sampleTimes = Range[0, 0.04, 10^-6];

peakField = Max[BzDipole /@ sampleTimes];
peakFlux = Max[flux /@ sampleTimes];
peakEMF = Max[emf /@ sampleTimes];

```

```

peakCurrent = Max[Icoil /@ sampleTimes];
peakVoltage = Max[Vload /@ sampleTimes];
peakTorque1 = Max[Abs[tauDipoleCoil /@ sampleTimes]];
peakTorque3 = Max[Abs[tauBiotSavart /@ sampleTimes]];

(*=====*)
(*PAPER VALUES*)
(*=====*)
paperField = 0.081;
paperFlux = 0.01018;
paperEMF = 3.20;
paperVrms = 2.26;
paperIrms = 0.215;
paperPower = 0.462;
paperTorque = 0.00147;

(*=====*)
(*ERROR FUNCTION*)
(*=====*)
err[x_, y_] := 100 Abs[(x - y) / y];

(*=====*)
(*RESULTS TABLE*)
(*=====*)
results = Grid[{{"Quantity", "Paper", "Simulation", "Error %"},
  {"Peak B (T)", paperField, peakField, err[peakField, paperField]},
  {"Peak Flux (Wb)", paperFlux, peakFlux, err[peakFlux, paperFlux]},
  {"Peak EMF (V)", paperEMF, peakEMF, err[peakEMF, paperEMF]},
  {"Vrms (V)", paperVrms, Vrms, err[Vrms, paperVrms]},
  {"Irms (A)", paperIrms, Irms, err[Irms, paperIrms]},
  {"Power (W)", paperPower, averagePower, err[averagePower, paperPower]},
  {"P/ω Torque (Nm)", paperTorque, torqueFromPower, err[torqueFromPower, paperTorque]},
  {"Dipole-Coil Torque (Nm)", paperTorque, avgTorque1Magnitude,
    err[avgTorque1Magnitude, paperTorque]}, {"Biot-Savart Torque (Nm)", paperTorque,
    avgTorque3Magnitude, err[avgTorque3Magnitude, paperTorque]}}, Frame → All];

Print[results];

(*=====*)
(*PLOTS*)
(*=====*)
p1 = Plot[BzDipole[t], {t, 0, 0.04}, PlotRange → All, GridLines → Automatic,
  AxesLabel → {"t (s)", "B_z (T)"}, PlotStyle → Blue, ImageSize → 300];

p2 = Plot[flux[t], {t, 0, 0.04}, PlotRange → All, GridLines → Automatic,
  AxesLabel → {"t (s)", "Flux (Wb)"}, PlotStyle → Red, ImageSize → 300];

```

```

p3 = Plot[emf[t], {t, 0, 0.04}, PlotRange → All, GridLines → Automatic,
  AxesLabel → {"t (s)", "EMF (V)"}, PlotStyle → Green, ImageSize → 300];

p4 = Plot[Icoil[t], {t, 0, 0.20}, PlotRange → All, GridLines → Automatic,
  AxesLabel → {"t (s)", "Current (A)"}, PlotStyle → Magenta, ImageSize → 300];

p5 = Plot[PowerOut[t], {t, 0.10, 0.20}, PlotRange → All, GridLines → Automatic,
  AxesLabel → {"t (s)", "Power (W)"}, PlotStyle → Black, ImageSize → 300];

p6 = Plot[{tauDipoleCoil[t], tauBiotSavart[t]}, {t, 0, 0.04}, PlotRange → All,
  GridLines → Automatic, AxesLabel → {"t (s)", "Torque (Nm)"}, PlotStyle → {Orange, Red},
  PlotLegends → {"Dipole-Coil", "Biot-Savart"}, ImageSize → 300];

Print[GraphicsGrid[{{p1, p2}, {p3, p4}, {p5, p6}}, ImageSize → Large]];

(*=====*)
(*FINAL NUMERIC OUTPUT*)
(*=====*)
Print["\n===== FINAL NUMERIC OUTPUT ====="];
Print["PeakField(T) = ", peakField];
Print["PeakFlux(Wb) = ", peakFlux];
Print["PeakEMF (V) = ", peakEMF];
Print["PeakVoltage (V) = ", peakVoltage];
Print["PeakCurrent (A) = ", peakCurrent];
Print["Vrms (V) = ", Vrms];
Print["Irms (A) = ", Irms];
Print["AveragePower (W) = ", averagePower];
Print["P/ω Torque (Nm) = ", torqueFromPower];
Print["Dipole-Coil Torque (Nm) = ", avgTorque1Magnitude, " (APPROXIMATE)"];
Print["Biot-Savart Torque (Nm) = ", avgTorque3Magnitude, " (MORE ACCURATE)"];

(*=====*)
(*RMS VOLTAGE ANALYSIS*)
(*=====*)
Print["\n===== RMS VOLTAGE ANALYSIS ====="];
VrmsIdeal = peakEMF / Sqrt[2];
Print["Ideal Vrms (peak/Sqrt[2]) = ", VrmsIdeal, " V"];
Print["Actual Vrms (from circuit) = ", Vrms, " V"];
Print["Reduction = ", (1 - Vrms / VrmsIdeal) * 100, "%"];
x1 = omega * Lcoil;
z = Sqrt[Rtotal^2 + x1^2];
Print["Impedance Z = ", z, " Ohm"];
phi = ArcTan[x1 / Rtotal];
Print["Phase angle φ = ", phi * 180 / Pi, " degrees"];

(*=====*)
(*ENERGY CONSERVATION CHECK*)

```

```

(*=====*)
Print["\n===== ENERGY CONSERVATION ====="];
mechPowerIn1 = avgTorque1Magnitude * omega;
mechPowerIn3 = avgTorque3Magnitude * omega;
Print["Electrical power out = ", averagePower, " W"];
Print["Mechanical power in (using Biot-Savart) = ", mechPowerIn3, " W"];
Print["Ratio = ", averagePower / mechPowerIn3];
If[Abs[averagePower - mechPowerIn3] < 0.01,
  Print["✓ Energy is approximately conserved!"],
  Print["Difference = ", averagePower - mechPowerIn3, " W"]];
B0 = 0.0812 T
Peak Bz = 0.0812 T

===== METHOD 1: DIPOLE-COIL INTERACTION =====
This is an APPROXIMATE electromagnetic reaction torque
Assumptions:
  1. Coil is treated as a point dipole
  2. Only on-axis field components are considered
  3. No integration over coil geometry
Average Dipole-Coil Torque = 0.00154517 Nm

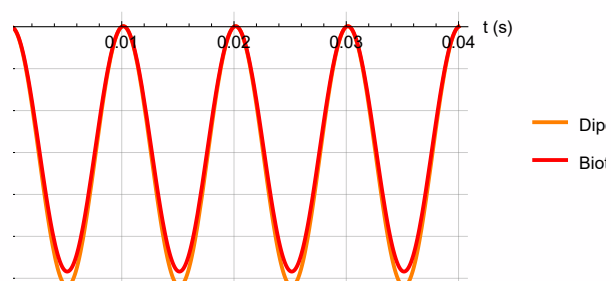
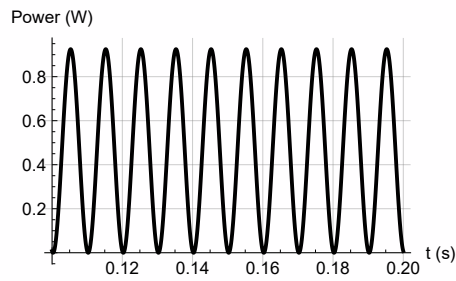
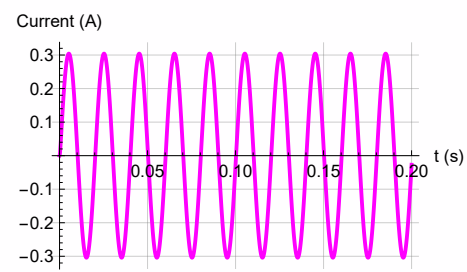
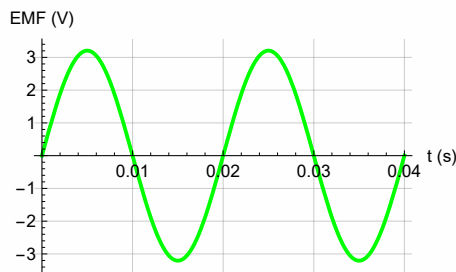
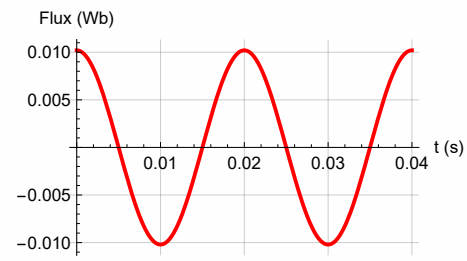
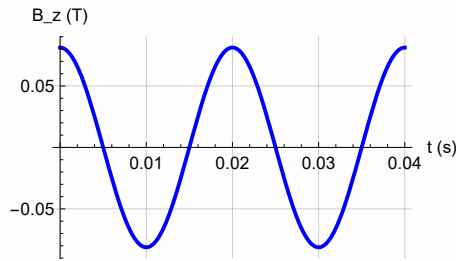
===== METHOD 2: TORQUE FROM POWER =====
Torque =  $P/\omega$  = 0.00147159 Nm
This is the mechanical torque required to generate P

===== METHOD 3: BIOT-SAVART TORQUE =====
This uses the actual field from a circular coil
Formula:  $B_{\text{coil}} = (\mu_0 * I * N * R^2) / (2 * (R^2 + z^2)^{3/2})$ 
Average Biot-Savart Torque = 0.00145689 Nm

===== COMPARISON OF TORQUE CALCULATIONS =====
Method 1 (Dipole-Coil Approximation): 0.00154517 Nm
Method 2 ( $P/\omega$ ): 0.00147159 Nm
Method 3 (Biot-Savart): 0.00145689 Nm
Ratio Method1/Method3: 1.0606
Ratio Method2/Method3: 1.01009

```

Quantity	Paper	Simulation	Error %
Peak B (T)	0.081	0.0812	0.246914
Peak Flux (Wb)	0.01018	0.0102039	0.234705
Peak EMF (V)	3.2	3.20565	0.176485
Vrms (V)	2.26	2.15015	4.86058
Irms (A)	0.215	0.215015	0.00702344
Power (W)	0.462	0.462315	0.0681675
P/ω Torque (Nm)	0.00147	0.00147159	0.108445
Dipole-Coil Torque (Nm)	0.00147	0.00154517	5.11387
Biot-Savart Torque (Nm)	0.00147	0.00145689	0.891705



===== FINAL NUMERIC OUTPUT =====

```

PeakField(T)           = 0.0812
PeakFlux(Wb)           = 0.0102039
PeakEMF(V)             = 3.20565
PeakVoltage(V)         = 3.04077
PeakCurrent(A)         = 0.304077
Vrms(V)                = 2.15015
Irms(A)                = 0.215015
AveragePower(W)        = 0.462315

```

P/ω Torque (Nm) = 0.00147159
 Dipole-Coil Torque (Nm) = 0.00154517 (APPROXIMATE)
 Biot-Savart Torque (Nm) = 0.00145689 (MORE ACCURATE)

===== RMS VOLTAGE ANALYSIS =====

Ideal Vrms (peak/Sqrt[2]) = 2.26674 V
 Actual Vrms (from circuit) = 2.15015 V
 Reduction = 5.14326%
 Impedance Z = 10.5422 Ohm
 Phase angle φ = 5.12911 degrees

===== ENERGY CONSERVATION =====

Electrical power out = 0.462315 W
 Mechanical power in (using Biot-Savart) = 0.457696 W
 Ratio = 1.01009
 ✓ Energy is approximately conserved!